Fall 2026

EDS 6352 – Natural Language Processing

Assignment #1 – Sentiment Analysis Using Naïve Bayes Text Classification

Use of artificial intelligence assistant such as ChatGPT in developing the code for this assignment is allowed and encouraged. However, the report needs to be completely written by you. Use of grammar correction tools is ***disallowed****.*

Assignment Overview

The objective of this assignment is to familiarize you with Naïve Bayes text (sequence) classifier as well as Spacy Python package for NLP. You will train a Naïve Bayes classifier to categorize some short movie reviews into two classes; positive and negative.

Assignment Description

Make sure you follow the instructions below very carefully. Do not use Gauss NB or Multinomial NB functions within the scikit-learn python package. You are expected to write your own Naïve Bayes classifier from scratch.

Input Data Description

You need to download the data from Kaggle at the following web address:
https://www.kaggle.com/datasets/yasserh/imdb-movie-ratings-sentiment-analysis

Analysis Required

Your objective is to write a code in Python (in a Jupyter notebook) to do the following:

- Read the movie reviews
- Sample and clean up the input (**see Data Sampling and Cleanup section below**) Note that you essentially are sampling training and test from a large dataset, so you do not get to use scikit-learn's train-test-split function for this assignment.
- Use regular expression (re) package to remove special characters and symbols (**See preprocessing section below**)
- Train a Naïve Bayes classifier that categorizes the reviews into two classes; positive and negative considering the following:
    - Calculate prior probabilities for each class

    - Calculate likelihood values of each word given each class
    - Calculate the posterior probability using logarithmic representation of probabilities
    - Use Laplace Smoothing (add 1)
    - Remove unknown words you encounter in the test data
- Perform sentiment analysis on the test data
- Calculate the confusion matrix and report precision, recall, and F-1 score for each class.

Data Sampling and Cleanup

Once you download the data (filename is movie.csv), load it in Excel and have a look at the data itself. The first column is "text" which is the movie reviews and the second column is "label" which is 0 for negative sentiment and 1 for positive. This file contains over 39,000 brief movie reviews. You need to randomly sample out 1000 reviews for training and 400 for testing. The training data should be sampled out completely randomly and without replacement. Once the training data is sampled out and removed from the file, you randomly sample out your 400 test data points. Make sure half of **test data** is from class 0 and half are from class 1.

Note: Make sure to look for missing/Null values in both columns before sampling and remove them if any.

Preprocessing

You need to create a preprocessing function that takes care of removing symbols, punctuations, extra spaces, etc. This function will look like this:

```
preprocess_text(text, lemmatize_words=True, remove_stop_words=True, handle_logical_negation=True)
```

Data cleanup should include removing special symbols such as <s>, </s>,
, <br\>, etc. But make sure that you replace them with proper punctuation such as "end of sentence period" because they help Spacy specially with logical negation.

Do not remove any punctuations from the input texts. Spacy requires all punctuations to be preserved to work best.

The three Boolean input arguments in your function preprocess_text, will help you run a few scenarios:

- Test if lemmatization helps improve the accuracy (per-class P, R, and F1).
- Test if removing stop words helps improve the accuracy (per-class P, R, and F1).
- Test if logical negation improves the accuracy metrics. Note that for logical negation handling to work, you need to lemmatize the input text (Spacy does this automatically when you create a doc object). So can't have lemmatize_words=Flase but handle_logical_negation=True.

Therefore, you need to run the entire training and testing 4 times:

1- Without lemmatization, removal of stop words, or handling logical negation
2- With lemmatization only

3- With lemmatization and removal of stop words
4- With lemmatization, removal of stop words, and handling logical negation

Note: You need to submit the accuracy assessment results for all 4 scenarios above. Accuracy assessment is done in form of reporting per-class precision, per-class recall, and per-class F1 score.

Tip: try to write your code with functions. Then you can dedicate one cell to calling appropriate pre-processing, training, testing, and accuracy assessment functions for each scenario.

Questions

Dedicate a section of your report to directly answering the following questions:

1- Compare and discuss the per-class precision, recall, and F1 values for each of the 4 scenarios. Do you see any meaningful and significant improvement in these measures of accuracy for any scenario? What is your overall suggested preprocessing combination for sentiment analysis using Naïve Bayes?
2- Explain the effect of Laplace smoothing. What would happen if you had not used it?
3- Based on your results, identify and discuss limitations of Naïve Bayes classifier. This involves looking at the text of misclassified movie reviews to identify patterns of shortcomings of the classifier and/or its limitations.

What to submit:

1- Your Python Jupyter Notebook should include plenty of comments and explanatory cells to ensure that your code is easy to read and understand. Make sure your code expects a file named movie.csv at the same path as the Jupyter notebook so it can be read without specifying the path. Also make sure you are not uploading the csv file alongside your Jupyter notebook. The Notebook should also contain the results of each cell executed by you. The notebook should be named **[Last_name]_HW1.ipynb**
2- A report (2-3 pages) **in PDF format** that briefly discusses the assignment and the results of 4 scenarios. Make one table that lists precision, recall, and f-1 score for each class and for each scenario. You can also compare overall accuracy values of the 4 scenarios. The last section of the report should clearly answer the questions asked in the questions section. The name of the file should be **[Last_name]_HW1.pdf**

Assignments are graded on a 100 scale.

Late submissions will be penalized at a rate of 1 point per hour.

We round up the time to the nearest hour.

Due Date: September 23, 2026 by 11:59 PM (submit through MS Teams)